Agenda.

- → Intro to Prompt Engineering
- → Elements Of Prompt Engineering
- → Parameters Of Prompt Engineering

Prompt Engineering is an art of writing a prompt with right set of words to guide the LLM's performance & getting better outputs.

→ Giving instructions to LLM to harness it's full potential & generate better responses.

Elements of a prompt.

- → Instruction
- → Context
- → Constraints
- → format
- → Variables.

Instruction.

↳ Task that we want LLM to perform.

Prompt : Generate a packing list for a trip.

Context.

↳ Some additional | background information about the task or instruction.

Prompt : Generate a packing list for a 5 days of business trip to Dubai from India in April.

Constraint

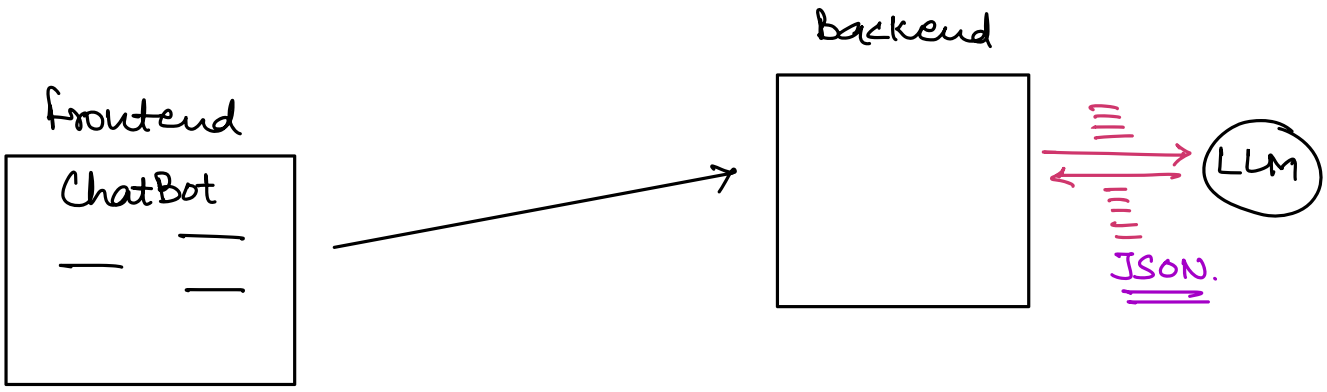
- Limitations | Restrictions we need to put on the model to generate the response
- What to include | What Not to include ?
- Length of the response
- Tone of the response.

Prompt : Generate a packing list for a 5 days of business trip to Dubai from India in April for 1 person with a luggage limit of 20KG.

Format.

↳ Structure or layout of the response generated by LLM.

→ How to organise & style the response.



Prompt : Generate a packing list for a 5 days of business trip to Dubai from India in April for 1 person with a luggage limit of 20KG, give the response in the email format.

```
{
  "id" : "____",
  "name" : "____",
  "age" : "_____",
  _____ : _____
  _____ : _____
}
```

Variables

↳ {variable_name}

{sender} = _____

{receiver} = _____

{destination} = _____

{no. of days} = _____

{purpose} = _____

_____

_____

_____

Prompt:

Write an email to {receiver} from {sender} with the packing list of items for trip to {destination} from {source} for {X} days - - - -

- - . . . -

Parameters of Prompt.

↳ Can be used to control / manage model's behaviour.

1) Temperature
2) Sampling
3) Repitition penalty
4) Max tokens.

Temperature.

→ Helps us to control the randomness / creativity of the output.

→ Range : 0 to 2

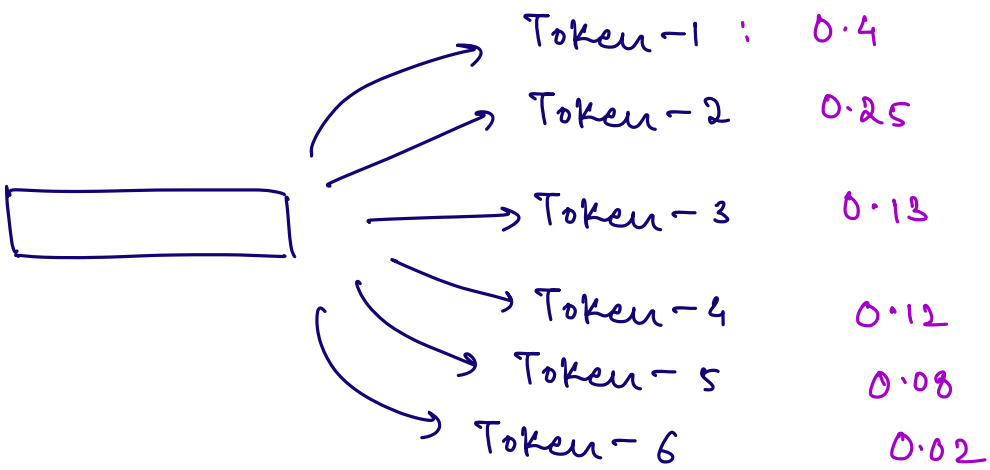
Low Temperature ⇒ More deterministic / predictible.

High Temperature ⇒ More randomness / creativity.

Temperature range	Use-cases | Examples.
0 – 0.4	Code | Algo | Math | reasoning..
0.4 – 0.8	Explanations
0.8 – 1.2	Posts | - - -
1.2 – 1.5	Story
> 1.5	Rarely used.

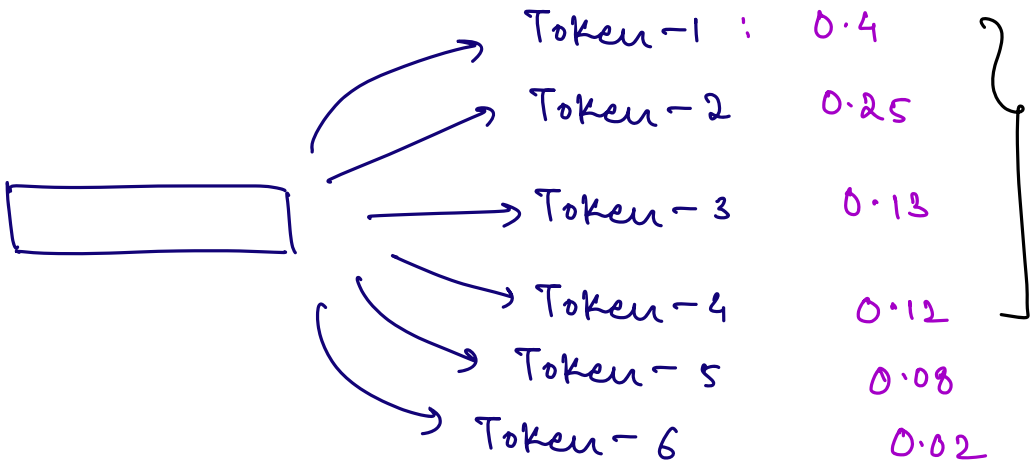
Sampling

↳ Controls how the model picks up the next tokens.



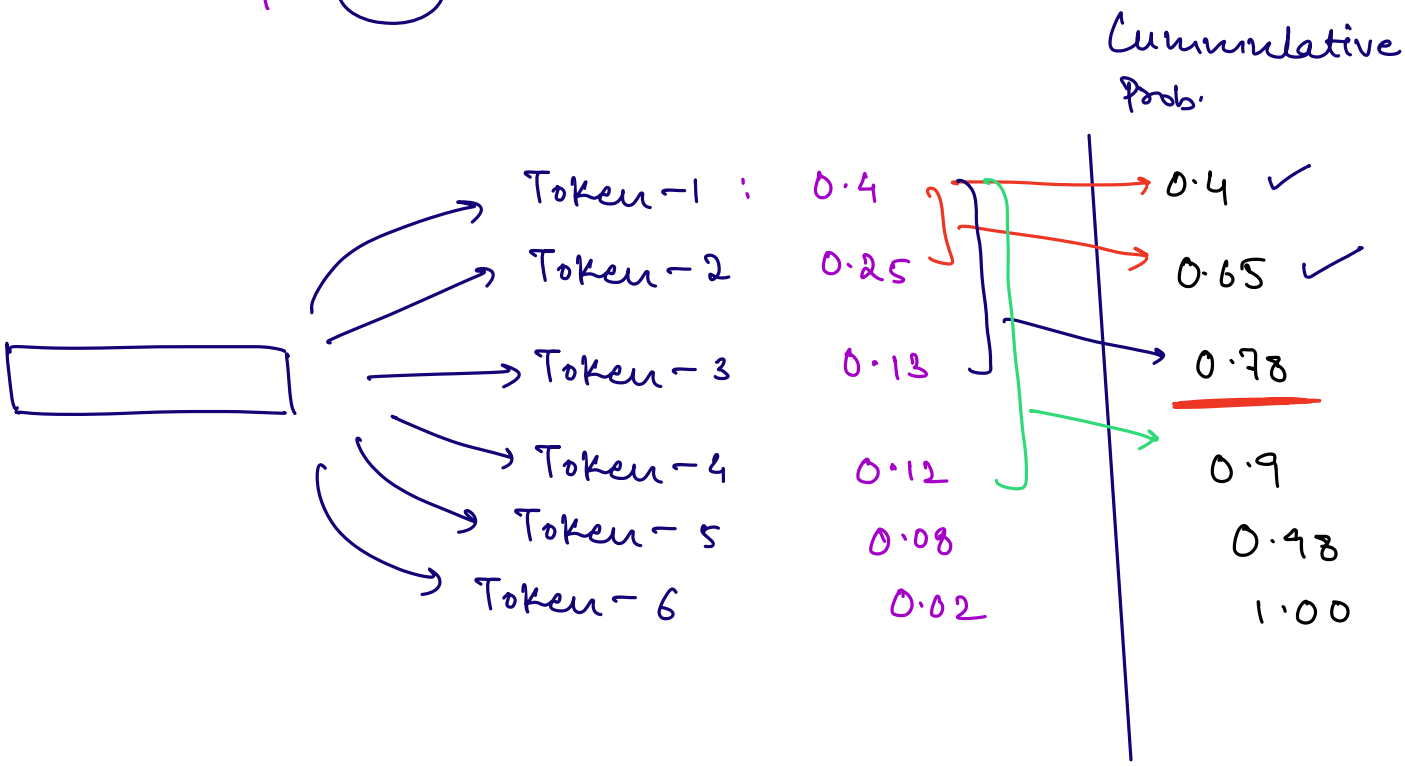
Top-k.

$k = 4$



Top-p.

$p = 0.7$



Repitition Penalty.

→ Controls the repitition of words or phrases the model has already generated.

→ Help to reduce redundancy.

Repitition Penalty.

1	→	No penalty
1 – 1.5	→	Mild repitition control.
1.5 – 1.7	→	Strong repitition control.
> 1.8	→	No repitition at all.

Max tokens.

↳ Maximum no. of tokens an LLM can process while generating one response.